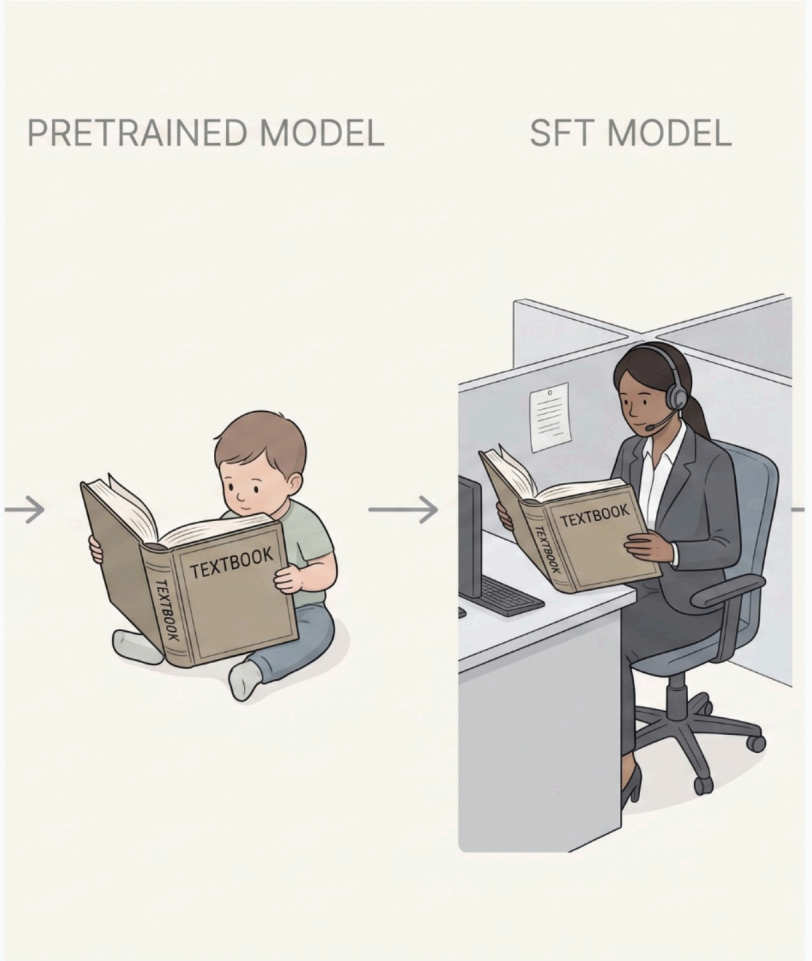
2. Supervised Finetuning (SFT)

SFT models produce text that is **nice to read**.

But they still haven't been trained to **reason over complex tasks**.



Reflection

1. Current models are excellent at computer programming and mathematics. What about these two domains allows humans to write high-quality responses?

2. What are some domains where it's not easy for humans to write high-quality responses?

3. Reinforcement Learning

SFT models may prioritize **“sounds nice”** over correctness.

Goal: teach a model to work on **long, difficult problems**.



Reinforcement Learning from Human Feedback

The model writes several responses; **a human picks the best one**.

Essay 1

“Within higher ed…”

Essay 2 — human’s pick

“The problems faced…”

Essay 3

“Students typically…”

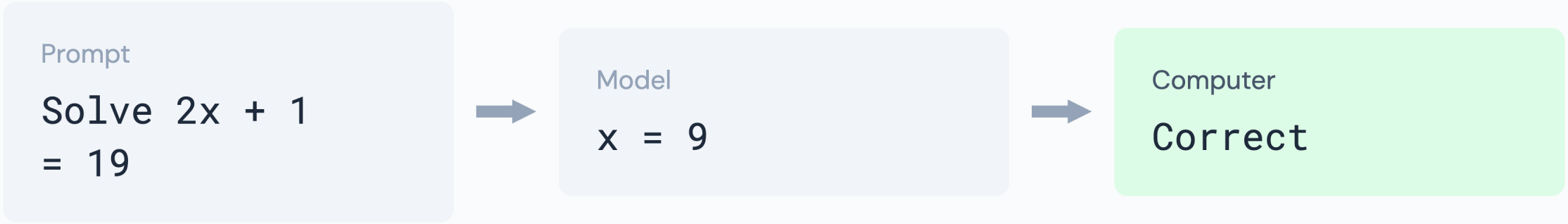
Going forward, the model prioritizes responses like essay 2.

Reflection

> RLHF is used less and less because it’s costly and brittle. Why might that be?

Reinforcement Learning from Verifiable Reward

A task with a verifiable solution. **A computer checks** the answer and rewards the model if it's correct.



Simple Idea, Enormous Development

1

Very short verifiable tasks

→

2

Progressively more complicated tasks

→

3

Tasks that take hours to complete

3. Reinforcement Learning

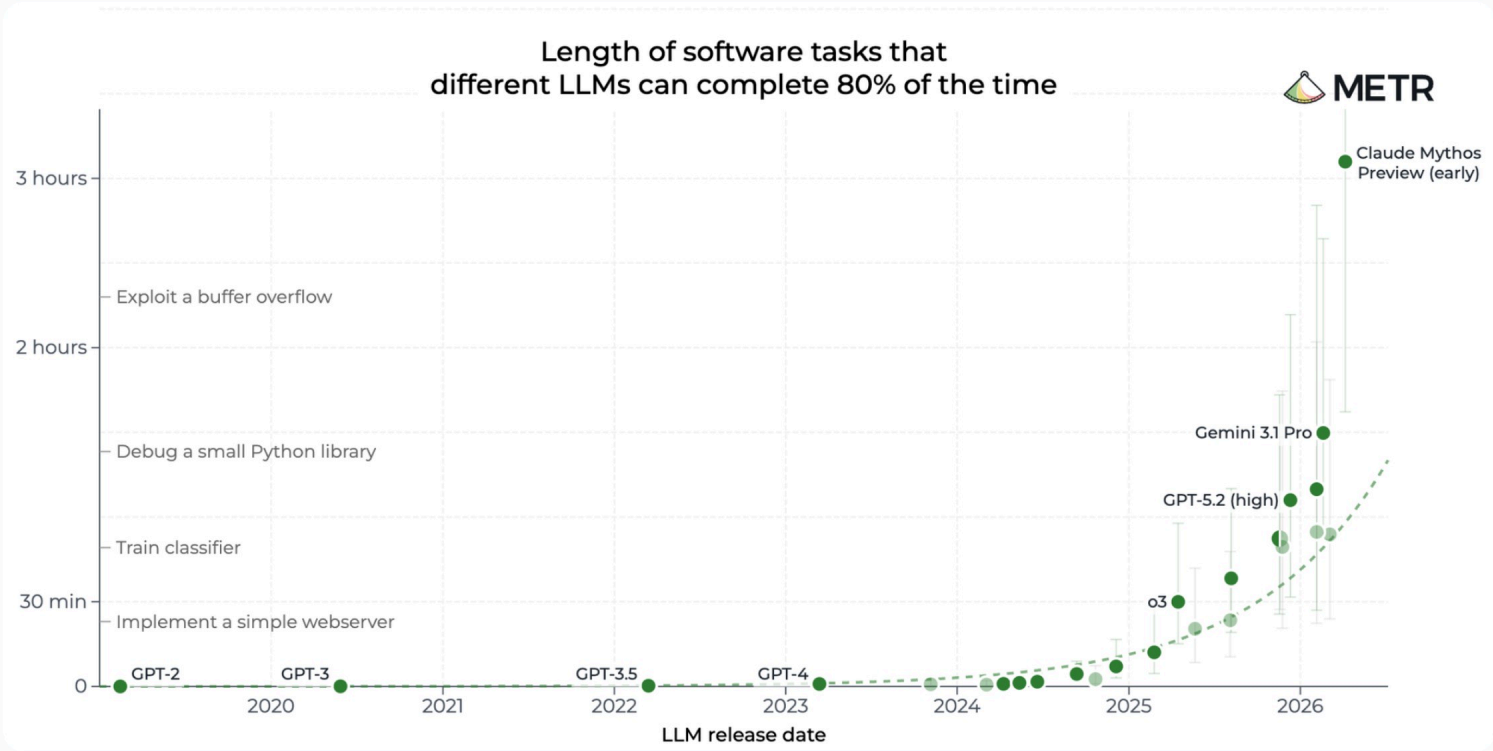
Pretraining: sees solutions to simpler problems

```
“To solve x² + 1 = 8, start by…”
```

Reinforcement learning: tries more difficult versions

```
“To solve 3x² + 2x + 1 = 8, I could start by…”
```

How Long Can Models Work?



Early models

A few seconds

2022

RLHF

2024

RLVR

A Few Caveats

Progress on coding and math is largely due to how easy those are for a computer to **verify**.

There's no reason to believe models will improve as fast in **other domains**.